

Phil Donovan

Data Architecture · Engineering · Science · AI

DIRECTOR – VIRTUS SOLUTIONS

PRODUCTS – [EOLAS.FYI](#)

I design data platforms — Snowflake, dbt, Iceberg, AWS — and the models that run on them, for councils, health agencies and insurers. I came to it from geography, after ten years of spatial analysis across public health, transport and urban policy.

phil@donovan.family · linkedin.com/in/phil-donovan-virtus-solutions · github.com/phildonovan · Auckland, New Zealand

Experience

2013 – PRESENT

2025 – now

Principal Consultant — Data Architecture, Engineering & AI

FIRM · AUCKLAND

- ▲ Led the design of Auckland Council's Modern Data Platform, centralising data assets for scale and governance.
- ▲ Built dbt, Snowflake and Fivetran pipelines — including automated ingestion of LINZ land parcels via WFS changesets, keeping council-wide spatial analysis current.
- ▲ Introduced and rolled out AI-assisted development tooling across the engineering team, setting the standards and workflows for its use.

2025 – now

Associate

PRINCIPAL ECONOMICS · AUCKLAND

- ▲ Built the Health Vulnerability and Adaptation dashboard for the Ministry of Health — an R Shiny and Plumber application modelling climate-change health impacts and adaptation options across New Zealand.
- ▲ Applied Treasury CBAX cost-benefit methodology, including declining discount rates, to cost adaptation scenarios down to SA2 level.
- ▲ Engineering on the Cost Intelligence Platform.

2024 – 2025

R/SQL Developer, Engineer & Scientist

AA INSURANCE · AUCKLAND

- ▲ Developed the pricing emulator that carried the migration from the legacy Protect platform to Duck Creek.
- ▲ Advised on the architecture of their AWS analysis platform, and built custom SageMaker container images so analysts were productive the moment one deployed.
- ▲ Wrote `aaair`, an R package joining Redshift and other sources to the pricing emulator for reproducible pricing simulation.

2023 – 2024	Senior Analyst, R Developer & SQL Engineer TE WHATU ORA – HEALTH NEW ZEALAND ▲ Led geospatial advisory for the Queenstown <i>Cryptosporidium</i> outbreak response, mapping contamination risk to direct public health intervention. ▲ Built <code>nphsintel</code>, an R package over Snowflake and PostGIS for population health analysis — cut code redundancy by around 75% and trained 10+ analysts on it. ▲ Introduced Quarto-based automated reporting and Shiny dashboards for spatial health trends.
2022 – 2023	Principal GIS Analyst MINISTRY OF HEALTH · WELLINGTON ▲ Automated geospatial workflows with the <code>mohanalytics</code> and <code>mohgeospatial</code> R packages, lifting team productivity by around 40%. ▲ Built and maintained the ArcGIS Online Portal in Python and R, surfacing interactive layers on health access disparities with ETL keeping them current. ▲ Wrote monitoring scripts tracking portal credit and cloud spend, cutting resource overruns by around 30%.
2018 – 2020	Spatial Data Scientist GEOGRAPHIC BUSINESS SOLUTIONS · AUCKLAND ▲ Deployed and configured the ESRI Enterprise stack — ArcGIS Pro and Server, Survey123, Collector, Operations Dashboard — for client geospatial platforms. ▲ Built 3D air pollution models for Seoul using the ArcGIS JavaScript API.
2016 – 2018	Spatial & Economic Data Analyst MRCAGNEY · AUCKLAND ▲ Delivered the National Policy Statement on Urban Development Capacity analysis — regression modelling of the effect of zoning regulation on land prices nationally. ▲ Ran the Whangārei residential and business development capacity assessment, building geometric algorithms to encode planning rules. ▲ Public transport modelling against the Auckland and Wellington transport models.
2013 – 2016	Spatial Data Analyst SHORE, MASSEY UNIVERSITY · AUCKLAND ▲ Principal spatial analyst on children's urban health research — from cleaning travel-diary data through to accessibility analysis. ▲ Co-authored five peer-reviewed papers on children's mobility and the built environment.

Capabilities	TOOLING
LANGUAGES	WAREHOUSES & ENGINES
R · Python · SQL	

Snowflake · Redshift · PostgreSQL · Oracle · SQL Server · DuckDB · Apache Arrow · Iceberg

PIPELINES

dbt · Meltano · Fivetran · Singer · PySpark

MODELLING

Regression · Supervised & unsupervised learning · Spatial models · Cost-benefit (CBAX)

APPS & REPORTING

R Shiny · Quarto · Power BI · ArcGIS Portal

CLOUD & INFRASTRUCTURE

AWS · Docker · Linux · Git

SPATIAL

PostGIS · ArcGIS · QGIS · FME · GRASS

Selected work

OPEN SOURCE & PROJECTS

BUILDING NOW

eolas

Official statistics for New Zealand, Australia and the OECD are scattered across dozens of portals, each with its own formats and quirks. eolas puts 700+ datasets behind a single typed API — clean columns, consistent schemas, no merged Excel headers.

I built the whole stack: Singer-spec taps and a Meltano orchestration layer landing data as Iceberg tables on S3 and Glue, exposed through a FastAPI service on AWS, with first-class Python and R clients, a CLI, and a Snowflake share for warehouse users.

PYTHON · R · SINGER · MELTANO · ICEBERG · FASTAPI · SNOWFLAKE · AWS

nzcensr

An R package that makes New Zealand census data straightforward to import, in both spatial and non-spatial form.

R · SPATIAL

NZ Economic Overview

An interactive Shiny dashboard over the vs-warehouse API, surfacing New Zealand economic indicators.

R · SHINY

NPS-UDC

National Policy Statement on Urban Development Capacity — quantifying how local government regulation moves land prices across New Zealand's major urban areas, by regression on zoning rules.

ECONOMICS · SPATIAL REGRESSION

NDAI-C

The Network Destination Accessibility Index for Children: a tool measuring how reachable everyday destinations are for kids across Auckland, Hamilton, Wellington and Christchurch.

ACCESSIBILITY · NETWORK ANALYSIS

nphsintel

An R package for Te Whatu Ora giving analysts one route into population health data across Oracle and Snowflake, with spatial analysis built in. Cut code redundancy by around 75%.

R · SNOWFLAKE · POSTGIS

Education

2007 — 2013

2012 — 2013

MSc in Geography — *First Class Honours*

University of Auckland

2011

PgDip in Geography

University of Auckland

2007 — 2010

BSc in Geography and Geology

Victoria University of Wellington

Published research

CO-AUTHORED

01 Kids in the City: Children's Use and Experiences of Urban Neighbourhoods in Auckland, New Zealand

Carroll, Witten, Kearns & Donovan

-
- 02 **Associations between the neighbourhood built environment and out of school physical activity and active travel**
Oliver, Mavoa, Badland, Parker, Donovan, Kearns, Lin & Witten
-
- 03 **Children's Out-of-School Independently Mobile Trips, Active Travel, and Physical Activity**
Oliver, Parker, Witten, Mavoa, Badland, Donovan, Chaudhury & Kearns
-
- 04 **Assessing neighbourhood destination access for children: development of the NDAI-C audit tool**
Badland, Donovan, Mavoa, Oliver, Chaudhury & Witten
-
- 05 **Development of a systems model to visualise the complexity of children's independent mobility**
Badland, Kearns, Carroll, Oliver, Mavoa, Donovan, Parker, Chaudhury, Lin & Witten
-

Get in touch

AUCKLAND • NZST

Open to conversations about data platform work, spatial architecture, and anything where the two meet.

EMAIL phil@donovan.family

LINKEDIN [phil-donovan-virtus-solutions](#)

GITHUB github.com/phildonovan

R-UNIVERSE phildonovan.r-universe.dev

PHIL DONOVAN — AUCKLAND, NEW ZEALAND